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(71) Applicant: **GKN AUTOMOTIVE LIMITED** [GB/GB];
2100 The Crescent Birmingham Business Park, Birmingham B37 7YE (GB).

(72) Inventors: **GARCÍA SIERRA, Iñigo**; c/o GKN Driveline Zumaia SA Sagarbidea 2, 20750 Zumaia (ES). **CABALLERO, Damián**; c/o GKN Driveline Zumaia SA Sagarbidea 2, 20750 Zumaia (ES). **TAMBURI, Damian Leandro**; 16 Balleneros Street, 20011 Donostia City (ES).

GARCÍA URBIETA, Jon; c/o GKN Driveline Zumaia SA Sagarbidea 2, 20750 Zumaia (ES). **PEÑA RODRIGUEZ, Alberto**; c/o GKN Driveline Zumaia SA Sagarbidea 2, 20750 Zumaia (ES).

(74) Agent: **NEUMANN MÜLLER OBERWALLENEY PATENTANWÄLTE PARTG MBB**; Overstolzstraße 2a, 50677 Köln (DE).

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(54) Title: ELECTRIC MACHINE

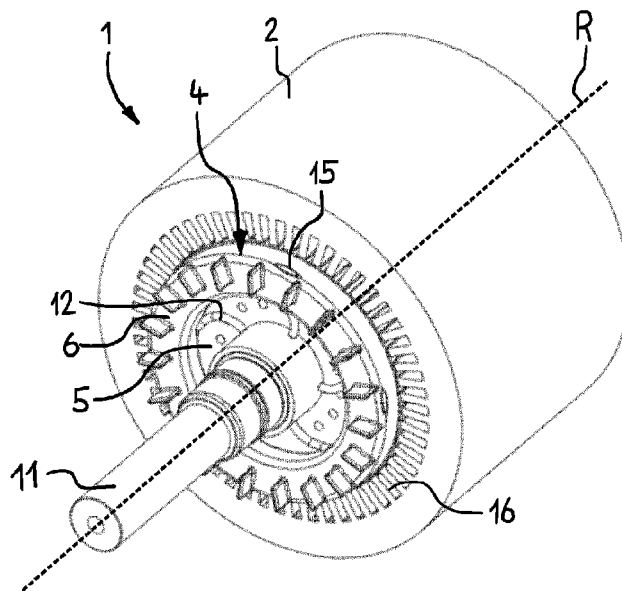


FIG. 2

(57) Abstract: Electric machine (1) comprising: a drive shaft (11) rotatable about an axis of rotation (L), said drive shaft (11) having an axial bore (14), a rotor (4) being non-rotatably secured to the drive shaft (11), said rotor (4) having a rotor core (5) and at least one rotor ring (6, 7) axially supported against an end of the rotor core (5), and at least one cooling channel (12) for conveying coolant, at least partially, in the at least one rotor ring (6, 7), wherein said at least one cooling channel (12) extends from an inlet opening (13), which is fluidly connected to the axial bore (14) of the drive shaft (11), to a radial outward outlet opening (15), characterised in that the at least one cooling channel (12) extends at least partially along a path that deviates from a straight radial direction.

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ZM, ZW.

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Electric Machine

Description

The invention relates to an electric machine comprising a drive shaft rotatable about an axis of rotation, and a rotor being non-rotatably secured to the drive shaft. Said drive shaft has an axial bore. Said rotor has a rotor core and at least one rotor ring axially supported against an end of the rotor core. At least one cooling channel for conveying coolant is provided, at least partially, in the at least one rotor ring, wherein said at least one cooling channel extends from an inlet opening, which is fluidly connected to the axial bore of the drive shaft, to a radial outward outlet opening.

A performance of an electric machine is thermally limited in operation. Inherent losses may occur in the copper, iron and magnets of electric machines, where material properties limit the temperatures of the respective components and structures. Effective cooling is necessary to achieve adequate performance. Especially for induction electric machines, such as asynchronous motors or generators, having a rotor with a squirrel cage, the hottest spot is the squirrel cage of the rotor.

JP 2009-118686 A discloses an electric machine as mentioned above with a cooling structure. The electric machine includes a stator and a rotor. The rotor is provided with a permanent magnet and the stator is provided with windings. For cooling, the electric machine has a first flow path for a coolant for cooling the permanent magnet and the windings. The electric machine has a second flow path for a coolant for cooling the windings, only. A flow-switching unit is provided that switches a flow state of the coolant to the first flow path and the second flow path based on one or both of the rotation speed and the output torque of the electric machine. Both flow paths are supplied with coolant from a supply bore in the centre of a drive shaft of the electric machine. The coolant is supplied to the flow paths by centrifugal forces generated by the rotation of

the drive shaft. Therefore, the first flow path and the second flow path are provided to penetrate linearly the hollow drive shaft from an inner peripheral surface to an outer peripheral surface along a radial direction of an axis of rotation. The first flow path continues into a rotor ring that is mounted onto the drive shaft and axially supported against a rotor core. The first flow path terminates in axial outlets in the rotor ring on axially opposite sides of the rotor ring to provide coolant to the rotor core and the windings of the stator.

US 11 018 555 B2 discloses an induction motor includes a motor shaft, a rotor, and a rotor conductor bar. The rotor has a first axial side and a second axial side and is non-rotatably secured to the motor shaft. The rotor includes a rotor core having an interior surface defining a slot extending from the first axial side to the second axial side. The rotor conductor bar is disposed in the slot. The rotor conductor bar and the slot cooperate to define a channel configured to transfer a fluid through the rotor core from the first axial side to the second axial side while directly contacting the rotor conductor bar. An inlet tube is coupled to the first axial side of the rotor and delivers the fluid to the channel. An outlet tube is coupled to the second axial side of the rotor extending outside of the rotor core and is configured to direct the fluid onto the windings of the stator.

The object of the invention is to enhance heat dissipation in the electric machine.

The objective is achieved by an electric machine according to claim 1. Exemplary embodiments are described in the dependent claims.

According to the invention, an electric machine comprises a drive shaft rotatable about an axis of rotation, and a rotor being non-rotatably secured to the drive shaft. Said drive shaft has an axial bore. Said rotor has a rotor core and at least one rotor ring axially supported against an end of the rotor core. At least one cooling channel for conveying coolant is provided, at least partially, in the at least one rotor ring, wherein said at least one cooling channel extends from an inlet opening, which is fluidly connected to the axial bore of the drive shaft, to a radial outward outlet opening. The at least one cooling channel extends at least partially along a path that deviates from a straight radial direction.

The path of the at least one cooling channel, therefore, deviates from a line extending from the axis of rotation in a radial direction. Accordingly, the cooling channel is arranged along a path that is longer than compared to a path in a straight radial direction. The longer path and therewith the increases heat exchange area enhances the heat dissipation from the at least one rotor ring to the coolant conveyed by centrifugal forces.

In order to connect the axial bore of the drive shaft with the at least one cooling channel, at least one supply channel can be provided in the drive shaft.

According to one exemplary embodiment, the inlet opening and the outlet opening of the at least one cooling channel are arranged offset to each other in a circumferential direction. The cooling channel, hence, extends both in a radial and in a circumferential direction. It is also conceivable that the inlet opening and the outlet opening are arranged on a radius that is a line segment extending from the axis of rotation in a straight radial direction, and that the cooling channel between the inlet opening and the outlet opening follows a path which is curve shaped, for example in a meandering form.

The circumferential offset of the inlet opening and the outlet opening may enclose an angle of more than 45 degrees, for instance of about 180 degrees. This means, that a radius, i.e. a line segment, from the axis of rotation to the inlet opening and a radius from the axis of rotation to the outlet opening enclose an angle of more than 45 degrees. The angle of the circumferential offset may be defined by the offset of the centre of the inlet opening and the centre of the outlet opening.

The inlet opening and the outlet opening of the at least one cooling channel may be arranged offset to each other in an axial direction. This offset of the openings will result in a longer path of the at least one cooling channel. In combination with the offset of the openings in a circumferential direction the length of the at least one cooling channel can be prolonged even further. The at least one cooling channel in this example would have a three-dimensional shape.

According to one exemplary embodiment, the outlet opening of the at least one cooling channel is arranged in an outer peripheral surface of said at least one rotor ring. The advantage of such a configuration is that in an outer area of the at least one rotor ring

the cooling channel extends as long as possible through the rotor ring and enhances heat dissipation. In addition, the coolant exits the outlet opening in a radial direction and splashes against the radially outward arranged components of the electric machine, such as windings of the stator.

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In particular, the outer peripheral surface is an outer cylindrical or conical surface of the at least one rotor ring.

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The at least one cooling channel may project out of an inner peripheral surface of the rotor ring towards the drive shaft. Alternatively, the inlet opening of the at least one cooling channel can be arranged in the inner peripheral surface of said at least one rotor ring.

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The at least one cooling channel can be formed, at least partially, in a curved shape around the axis of rotation, in order to prolong the length of the path of the at least one cooling channel.

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According to a specific embodiment, the at least one cooling channel is formed, at least partially, in a spiral shape around the axis of rotation. The at least one cooling channel can be made of different segments. The radius of each individual segment may increase gradually in direction towards the outlet opening, to accomplish the packaging of the component in terms of filling the rotor ring area and avoiding collisions between different cooling channels.

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For instance, the at least one cooling channel might start from the inlet opening in a first segment along a straight radial direction followed by a second segment along a section of a circle. The second segment merges into a third segment along a section of a circle, wherein the radius, i.e. the length of a line segment extending from the centre of the circle to the circumference of the circle, of the second section is smaller than the radius of the third section. The third segment might merge into a segment with a straight extension along a secant of the outer periphery of the at least one rotor ring. Between the second and third segments any number of further segments may be arranged, wherein the radius of the segments increases from the inside to the outside of the at least one rotor ring. In the context of this invention, such a configuration is still

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considered spiral.

According to an exemplary embodiment, the at least one cooling channel is a tube. The advantage of a cooling channel being a tube is, for instance, that the at least one cooling channel can be produced in a simple way in any conceivable shape. After the cooling channel has been formed into the desired form, a base material of the at least one rotor ring can be, at least partially, moulded over the tube. The tube may have a circular, rectangular or any other possible profile in a cross-sectional view.

10 In order to facilitate the over-moulding process, the tube can be made of a material that has a higher melting temperature than the base material of the at least one rotor ring. For instance, the base material of the at least one rotor ring can be aluminium, and the tube can be made of copper.

15 The electric machine can further comprise a stator having windings of an electrically conductive material, and the outlet opening of the at least one cooling channel can be arranged axially in overlap with axial ends of the windings. Thereby, not only the rotor is cooled efficiently, but also the ends of the stator windings.

20 These and other features and advantages of the invention will become apparent upon further reading of the specification in light of the accompanying drawings.

Figure 1 is a schematic longitudinal sectional view of an electric machine,

25 Figure 2 is a perspective view of the electric machine of Figure 1,

Figure 3 is a schematic front view of a rotor ring of the electric machine of Figure 1,

Figure 4 is a side view of the rotor ring of Figure 3,

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Figure 5 is a front view of a tube of one of the cooling channels of the rotor ring of Figures 3 and 4,

Figure 6 is a side view of the tube of Figure 5,

Figure 7 is a cross-sectional view of the electric machine of Figure 1 in the area of the rotor ring, and

5 Figure 8 is an enlarged partial view of Figure 7 showing the tube connected to a drive shaft of the electric machine.

Figure 1 shows a schematic longitudinal section of an asynchronous electric machine 1, in particular an asynchronous motor, with a stator 2. The stator 2 has windings 3
10 made of an electrically conductive material and connected to alternating current to induce a magnetic field rotating around an axis of rotation L. Figure 2 shows a perspective view of the electric machine 1 but does not depict the windings 3 but slots 16 that accommodate the windings 3. A rotor 4 is arranged within the stator 2 rotatable about the axis of rotation L. The rotor 4 has a magnetically conductive rotor core 5, which is
15 made as a laminated core or of sinter material. The rotor 4 has two rotor rings 6, 7 each axially supported against one axial end of the rotor core 5 so that the rotor core 5 is sandwiched between the two rotor rings 6, 7. The rotor rings 6, 7 are configured as short-circuit rings of a squirrel cage 8 of the rotor 4. The squirrel cage 8 comprises, besides the rotor rings 6, 7, several conductor bars 9, which are arranged in slots
20 of the rotor core 5 and which are connected to each of the rotor rings 6, 7. The squirrel cage 8 is made of an electric conductive material so that the rotating magnetic field produced by the stator 2 induces a current in the conductor bars 9, which are short-circuited by the rotor rings 6, 7. The induced current in the conductor bars 9 produce another magnetic field that lets the rotor rotate about the axis of rotation L. The squirrel
25 cage 8 can be built of separate rotor rings 6, 7 and separate conductor bars 9, which are connected to each other in an appropriate mechanical way, such as welding. Alternatively, the squirrel cage 8 can be die cast as one piece to the rotor core 5.

The rotor 4, in particular the rotor core 5 is secured to a drive shaft 11 in a non-rotatable
30 manner so that the torque generated by the rotating magnetic field is transferred from the rotor 4 to the drive shaft 11. It is understood that the asynchronous machine 1 can also be operated as electric generator so that by rotating the drive shaft 11 a rotating magnetic field is induced in the stator 2 for generating electrical power.

In the disclosed embodiment, the rotor rings 6, 7 are radially spaced apart from the drive shaft 11. It is, however, also possible that the rotor rings 6, 7 are seated on the drive shaft 11 in a non-rotatable manner in order to transfer torque.

5 The rotor rings 6, 7 are each provided with a plurality of cooling channels 12 for conveying coolant. The coolant can be any fluid feasible for extracting heat energy, such for instance a cooling oil. It is also possible that only one of the rotor rings 6, 7 is provided with such cooling channels 12. The rotor rings 6, 7 may also be provided with only one cooling channel 12.

10

The cooling channels 12 each extend from an inlet opening 13, which is fluidly connected to an axial bore 14 within the drive shaft 11 to a radial outward outlet opening 15. Hence, all cooling channels 12 are supplied with coolant from the axial bore 14 of the drive shaft 11 by centrifugal forces when the drive shaft 11 and the rotor rings 6, 7 rotate about the axis of rotation L.

15

The cooling channel 12 extend at least partially along a path that deviates from a straight radial direction. Radial direction means any direction along a radius that starts from the axis of rotation L and is arranged perpendicular to the axis of rotation L. The reason for that is to provide a larger length of the cooling channels 12 within the rotor rings 6, 7 compared to a cooling channel that extends in a straight radial direction. This results in a larger heat exchange area between the cooling channels 12 and the further material of the rotor rings 6, 7.

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25 Figure 3 depicts a front view of one of the rotor rings 6 with schematically shown cooling channels 12. The cooling channels 12 are illustrated with solid lines despite the fact that a large part of the cooling channels 12 is arranged within a surrounding base material 20 of the rotor ring 6.

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30 The rotor ring 6 comprises 5 cooling channels 12, which are evenly distributed around the axis of rotation L, one of which is provided with reference numerals and will be described hereinafter.

30

The cooling channels 12 are arranged over the greatest part of the length within the base material 20. The cooling channels 12 extend from an inner peripheral surface 17 to an outer peripheral surface 18 of the rotor ring 6. By means of conveying coolant through the cooling channels 12, the base material 20 and the entire rotor ring 6 can be cooled efficiently. In order to enhance the cooling effect, the rotor ring 6 is provided with fins 19, which project in an axial direction on a side opposite the rotor core 5 as can best be seen in Figure 1. The fins 19 are optional and the rotor ring 6 can also be configured without fins 19.

The cooling channels 12 are provided in the form of separate tubes, one of which is depicted in Figures 5 and 6. The cooling channels 12 in form of tubes project over the inner peripheral surface 17 inwardly. The inlet openings 13 of the cooling channels 12 are connected with the axial bore 14 of the drive shaft 11.

In the disclosed embodiment, the outlet openings 15 of the cooling channels 12 are flush with the outer peripheral surface 18 of the rotor ring 6. However, it is also conceivable that the cooling channels 12 in form of tubes project beyond the outer peripheral surface 18 outwardly.

In order to provide a long path of the cooling channels 12 within the base material 20 of the rotor ring 6, the inlet openings 13 and the outlet openings 15 of each cooling channel 12 are arranged offset to each other in a circumferential direction. In the disclosed embodiment, the circumferential offset angle α of the inlet opening 13 and the outlet opening 15 is almost 180 degrees. Accordingly, the cooling channel 12 almost extends over half the circumference of the rotor ring 6.

As can be seen in Figure 4, which shows a side view of the rotor ring 6, the inlet openings 13 and the outlet openings 15 of each of the cooling channels 12 are offset to each other in an axial direction. The offset is indicated with the reference sign S.

The outlet openings 15 of the cooling channels 12 are arranged axially in overlap with end of the windings 3, as can be seen in Figure 1. In response to the centrifugal forces that act on the coolant during rotation of the rotor 4, the coolant is ejected out of the

cooling channels 12 and splashes against the ends of the windings 3. Thereby, also the windings 3 are cooled.

The cooling channels 12 are formed in a curved shape around the axis of rotation L, in particular in a spiral or spiral-like shape as can be seen in Figure 5. Each cooling channel 12 has different segments. The radius of each individual segment increases gradually in direction towards the outlet opening 15.

The cooling channel 12 as depicted in Figure 5 starts from the inlet openings 13 with a first segment 21, which follows a straight line. The first segment 21 merges into a second segment 22, which follows a section of a circle having a first radius R_1 . A third segment 23 having a second radius R_2 follows the second segment 22. A following fourth segment 24 has a third radius R_3 . The final fifth segment 25 again follows a straight line.

The first radius R_1 is smaller than the second radius R_2 , and the second radius R_2 is smaller than the third radius R_3 . Between the different segments 21, 22, 23, 24, 25 further segments are arranged. The fifth segment 25 is arranged along a secant of the rotor ring 6. The rotor ring 6 is produced by an over-moulding process of base material 20 around the cooling channels 12 in the form of tubes. The melting temperature of the cooling channels 12 in the form of tubes is higher than that of the base material 20, so that during the over-moulding process of the cooling channels 12 are not affected. For instance, the base material 20 can be aluminium and the material of the cooling channels 12 in form of tubes can be copper.

The cooling channels 12 are connected to radial arranged supply channels 26, which lead from the axial bore 14 of the drive shaft 11 to an outer peripheral surface of the driveshaft 11. In the shown embodiment (see Figures 7 and 8), the first segment 21 of the cooling channels 12 is arranged within the supply channels 26. It is also conceivable that the cooling channels 12 only partially are arranged within the supply channels 26, or are connected to the radial outer end of the supply channels 26.

List of reference numerals

1	electric machine
2	stator
3	winding
4	rotor
5	rotor core
6	rotor ring
7	rotor ring
8	squirrel cage
9	conductor bar
10	slot
11	drive shaft
12	cooling channel
13	inlet opening
14	axial bore
15	outlet opening
16	slot
17	inner peripheral surface
18	outer peripheral surface
19	fin
20	base material
21	first segment
22	second segment
23	third segment
24	fourth segment
25	fifth segment
26	supply channel
L	axis of rotation
R ₁	first radius
R ₂	second radius
R ₃	third radius
S	offset in axial direction
α	offset in circumferential direction

Claims

1. Electric machine (1) comprising:
a drive shaft (11) rotatable about an axis of rotation (L), said drive shaft (11) having an axial bore (14),
a rotor (4) being non-rotatably secured to the drive shaft (11), said rotor (4) having a rotor core (5) and at least one rotor ring (6, 7) axially supported against an end of the rotor core (5), and
at least one cooling channel (12) for conveying coolant, at least partially, in the at least one rotor ring (6, 7), wherein said at least one cooling channel (12) extends from an inlet opening (13), which is fluidly connected to the axial bore (14) of the drive shaft (11), to a radial outward outlet opening (15),
characterised in
that the at least one cooling channel (12) extends at least partially along a path that deviates from a straight radial direction.
2. Electric machine (1) according to claim 1,
characterised in
that at least one supply channel (26) is provided in the drive shaft (11), and
that said supply channel (26) fluidly connects the axial bore (14) of the drive shaft (11) with the at least one cooling channel (12).
3. Electric machine (1) according to claim 1 or 2,
characterised in
that the inlet opening (13) and the outlet opening (15) of the at least one cooling channel (12) are arranged offset to each other in a circumferential direction.

4. Electric machine (1) according to claim 3,
characterised in
that a circumferential offset the inlet opening (13) and the outlet opening (15) encloses an angle (α) of more than 45 degrees.
5. Electric machine (1) according to one of claims 1 to 4,
characterised in
that the inlet opening (13) and the outlet opening (15) of the at least one cooling channel (12) are arranged offset to each other in an axial direction.
6. Electric machine (1) according to one of claims 1 to 5,
characterised in
that the outlet opening (15) of the at least one cooling channel (12) is arranged in an outer peripheral surface (18) of said at least one rotor ring (6, 7).
7. Electric machine (1) according to one of claims 1 to 6,
characterised in
that the at least one cooling channel (12) projects out of an inner peripheral surface (17) of the rotor ring (6, 7) towards the drive shaft (11).
8. Electric machine (1) according to one of claims 1 to 7,
characterised in
that the at least one cooling channel (12) is formed, at least partially, in a curved shape around the axis of rotation (L).
9. Electric machine (1) according to one of claims 1 to 8,
characterised in
that the at least one cooling channel (12) is formed, at least partially, in a spiral shape around the axis of rotation (L).
10. Electric machine (1) according to one of claims 1 to 9,
characterised in
that the at least one cooling channel (12) is a tube.

11. Electric machine (1) according to claim 10,
characterised in
that the at least one rotor ring (6, 7) comprises a base material (20) that is, at least partially, moulded over the tube.
12. Electric machine (1) according to claim 11,
characterised in
that the tube is made of a material that has a higher melting temperature than the base material (20) of the at least one rotor ring (6, 7).
13. Electric machine (1) according to claim 11 or 12,
characterised in
that the base material (20) of the at least one rotor ring (6, 7) is aluminium, and the tube is made of copper.
14. Electric machine (1) according to one of claims 1 to 13,
characterised in
that the electric machine (1) further comprises a stator (2) having windings (3) of an electrically conductive material and
that the outlet opening (15) of the at least one cooling channel (12) is arranged axially in overlap with axial ends of the windings (3).

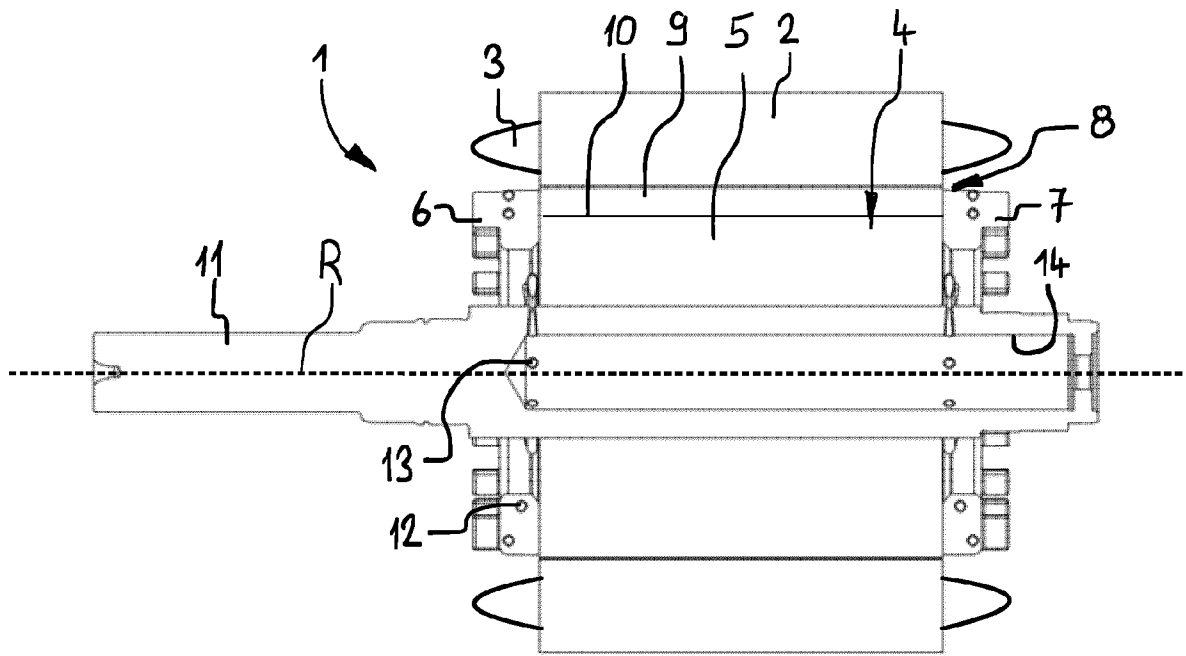


FIG. 1

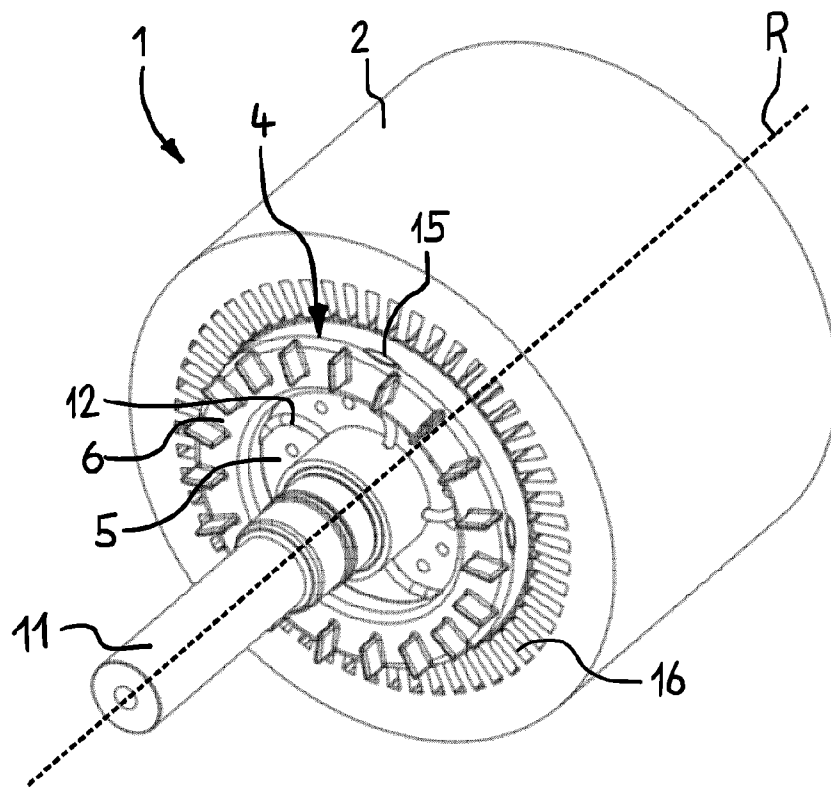
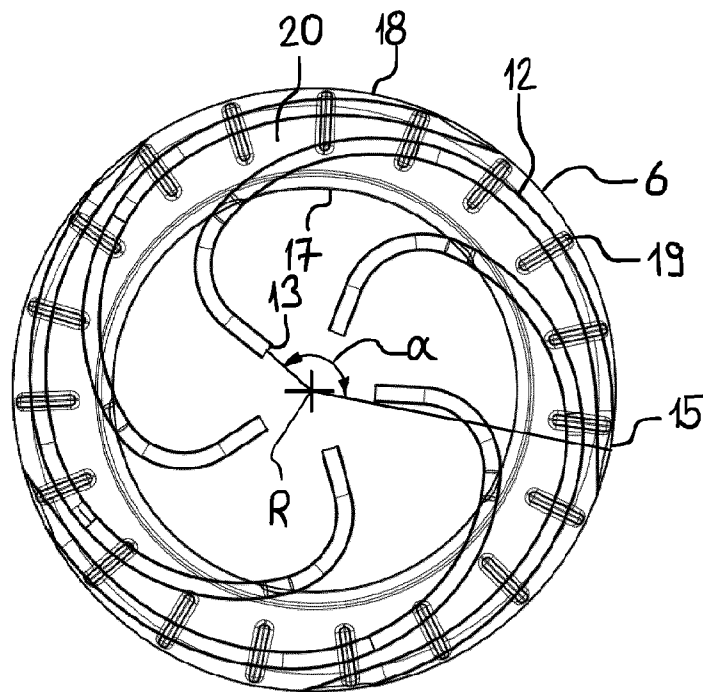
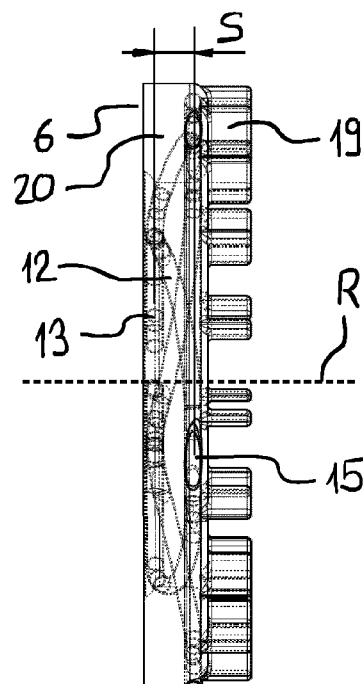


FIG. 2

**FIG. 3****FIG. 4**

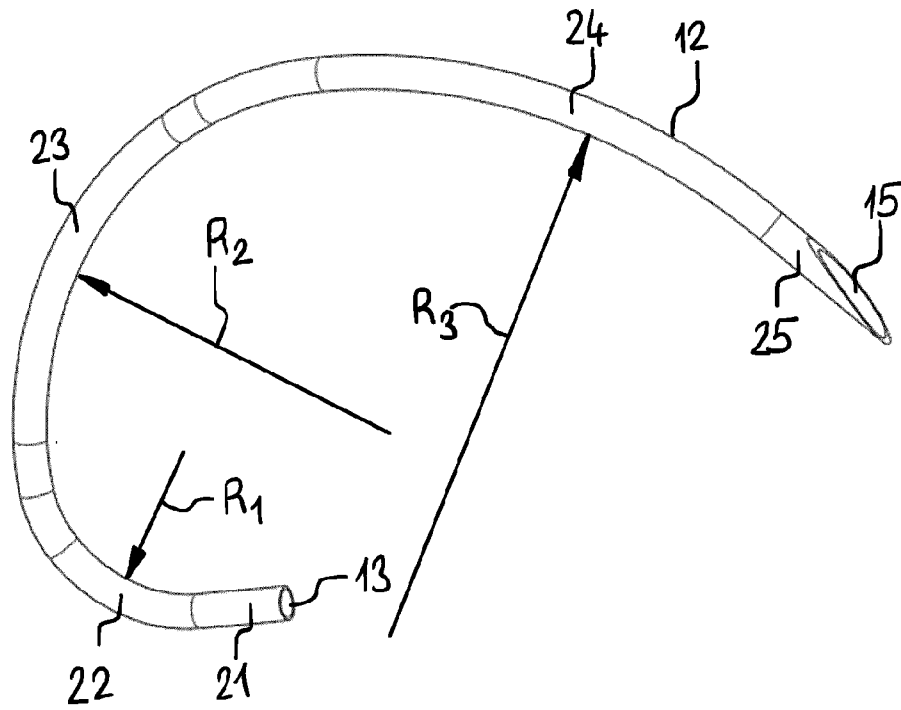


FIG. 5

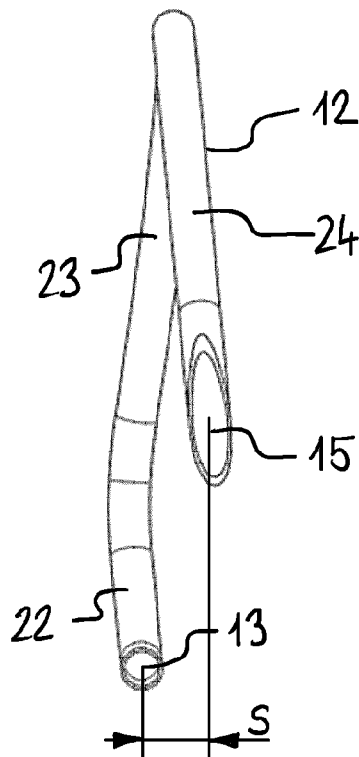


FIG. 6

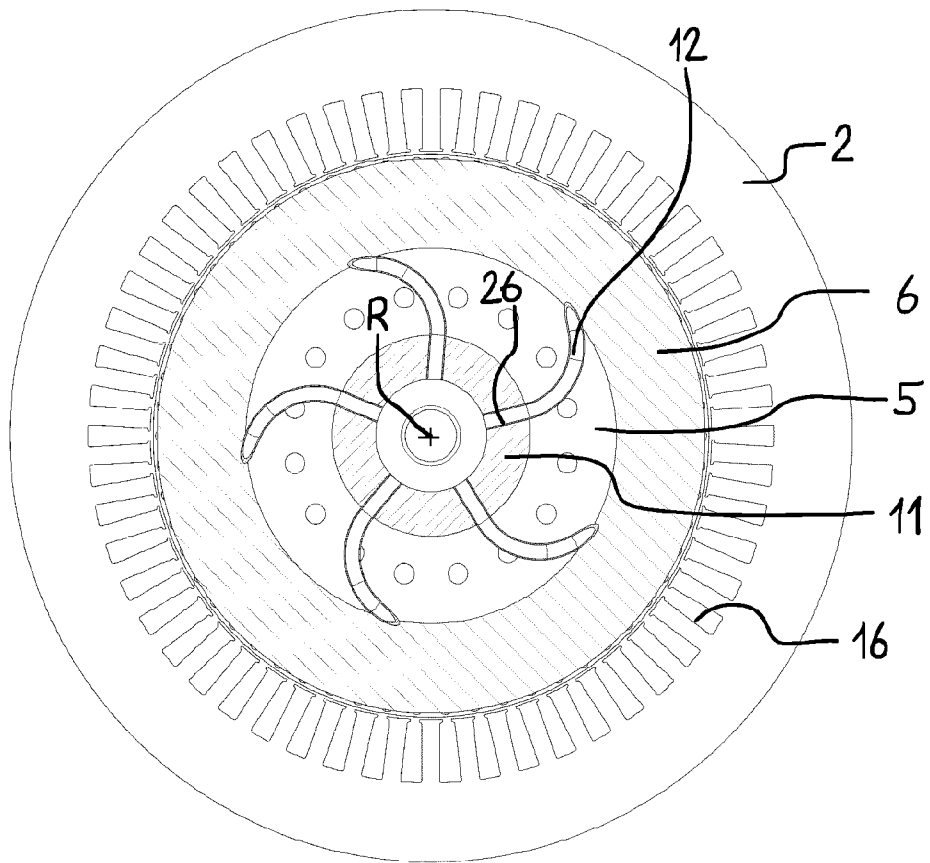


FIG. 7

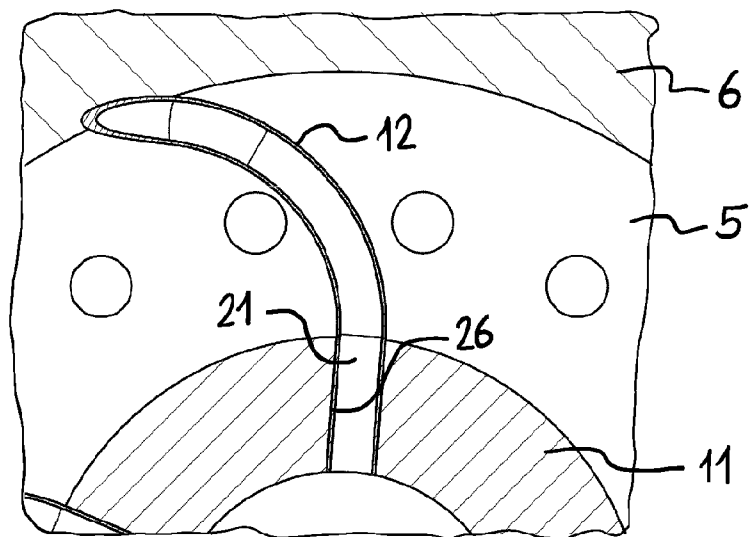


FIG. 8

INTERNATIONAL SEARCH REPORT

International application No

PCT/EP2023/050211

A. CLASSIFICATION OF SUBJECT MATTER

INV. H02K1/32 H02K9/19
ADD. H02K7/00

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

H02K

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPO-Internal, WPI Data

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	JP 2011 087434 A (TOYOTA MOTOR CORP) 28 April 2011 (2011-04-28)	1-5, 7, 8
Y	figures 1, 2, 4 -----	9-13
X	US 2019/173344 A1 (ISHIKAWA YUKI [JP] ET AL) 6 June 2019 (2019-06-06) figure 9 -----	1
X	FR 3 111 029 A1 (NOVARES FRANCE [FR]) 3 December 2021 (2021-12-03) figures 1, 4, 6 -----	1, 6
X	WO 2022/102219 A1 (HITACHI ASTEMO LTD [JP]) 19 May 2022 (2022-05-19) figures 1, 4 ----- -/-	1, 14



Further documents are listed in the continuation of Box C.



See patent family annex.

* Special categories of cited documents :

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Date of the actual completion of the international search

4 July 2023

Date of mailing of the international search report

12/07/2023

Name and mailing address of the ISA/

European Patent Office, P.B. 5818 Patentlaan 2
NL - 2280 HV Rijswijk
Tel. (+31-70) 340-2040,
Fax: (+31-70) 340-3016

Authorized officer

Jabri, Tarak

INTERNATIONAL SEARCH REPORT

International application No

PCT/EP2023/050211

C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT		
Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
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